

MongoDB User Roles Practical

Database: insta_clone

1. Create the insta_clone Database

Command:

```
mongosh
use insta_clone
```

Expected Output:

```
switched to db insta_clone
```

Explanation: MongoDB selects the insta_clone database. The database is persisted when a database object, such as a user or collection, is created.

2. Create storyAdmin User with dbAdmin Role

Command:

```
db.createUser({
  user: "storyAdmin",
  pwd: "StoryAdmin@123",
  roles: [
    { role: "dbAdmin", db: "insta_clone" }
  ]
})
```

Expected Output:

```
Successfully added user:
{
  "user" : "storyAdmin",
  "roles" : [
    {
      "role" : "dbAdmin",
      "db" : "insta_clone"
    }
  ]
}
```

Explanation: The storyAdmin user receives the dbAdmin role for database administration.

3. Create Read-Only feedViewer User

Command:

```
db.createUser({
  user: "feedViewer",
  pwd: "FeedViewer@123",
  roles: [
    { role: "read", db: "insta_clone" }
  ]
})
```

Expected Output:

```
Successfully added user:
{
  "user" : "feedViewer",
  "roles" : [
```

```

    {
      "role" : "read",
      "db" : "insta_clone"
    }
  ]
}

```

Explanation: The feedViewer user can read data but cannot insert, update, or delete documents.

4. Authenticate as feedViewer and Attempt an Insert

Command:

```

Exit the current shell:
exit

Authenticate:
mongosh -u "feedViewer" -p "FeedViewer@123" --authenticationDatabase "insta_clone"

use insta_clone

db.posts.insertOne({
  username: "aruna",
  caption: "My first Instagram post"
})

```

Expected Output:

```

MongoServerError: not authorized on insta_clone to execute command
{ insert: "posts", documents: [ ... ] }

```

Explanation: The insert fails because feedViewer has only the read role. The read role permits queries such as `db.posts.find()`, but does not grant write permissions such as `insertOne()`, `updateOne()`, or `deleteOne()`.

5. List All Users

Command:

```
db.getUsers()
```

Expected Output:

```

{
  "users" : [
    {
      "user" : "storyAdmin",
      "db" : "insta_clone",
      "roles" : [
        {
          "role" : "dbAdmin",
          "db" : "insta_clone"
        }
      ]
    },
    {
      "user" : "feedViewer",
      "db" : "insta_clone",
      "roles" : [
        {
          "role" : "read",
          "db" : "insta_clone"
        }
      ]
    }
  ],
  "ok" : 1
}

```

}

Explanation: Role difference: read allows viewing/querying database data only. dbAdmin provides database-administration privileges such as managing database metadata and indexes; it is intended for database management.

Role Summary

User	Role	Purpose
storyAdmin	dbAdmin	Database administration and management
feedViewer	read	Read/view database data only

Final Result: storyAdmin is assigned dbAdmin, while feedViewer is assigned read. The attempted insert by feedViewer is rejected with an authorization error because the user has no write privileges.